

JobCore: AI Agent To Get Your Job

Project Proposal and Review

CSE499B | Section - 12 | Project Group - 03

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Introduction

Searching for a job is still a fragmented and time-consuming process for candidates, with the hassle of multiple online applications, multiple postings, and minimal interview preparation. This project outlines the design and development of an Artificial Intelligence-enabled agent that will improve and streamline the whole job application process, from job searching all the way through interview preparation.

The suggested system combines various job portals, including LinkedIn, to aggregate and expose opportunities related to user-defined job positions and skills. Using sophisticated natural language processing and machine learning algorithms, the agent intelligently matches jobs, presenting granular job listings based on individual candidate profiles. The system includes an auto-resume analysis module that recommends improvements before applications are submitted, as well as intelligent form-filling features that automatically fill out and submit a job application on behalf of a user using previously entered user information. In addition to application management, the assistant fully supports students' interview preparation with curated YouTube and other educational platform learning resources. One of the advancements that emerged is implementing an AI-driven mock interview system that utilizes voice interaction technology to generate performance reports with extensive personal feedback. The system also includes a company-specific interview process feature and can integrate with Google Calendar for interviews.

Problem Statement

- Users manually search for jobs across many platforms.
- Applications require repeating the same information multiple times.
- Users rarely get real feedback on resumes or interviews.
- Interview preparation resources are scattered.

Objectives

- Build a smart job searching system using AI matching.
- Automatically optimize the user's resume according to the job description.
- Create an AI mock interview with scoring and feedback.
- Automate application submission.
- Develop custom models rather than relying entirely on APIs.
- Compare multiple LLM models for performance.

System Features

1. User Authentication: signup and login
2. Dashboard: manage profile and track progress
3. AI Job Search: skill-based job recommendation
4. Resume Optimization: ATS-friendly rewrite suggestions
5. Automated Application: autofill application forms
6. AI Mock Interview: text and voice-based interview

AI Models and Datasets

Models: LLAMA, DeepSeek, Qwen, Mistral, Granite, Claude

Interview Dataset:

Software Engineering Interview Questions Dataset

- Size: 250 questions
- Topics: Algorithms, System Design, ML, Data Structures, Networking, Security, Databases, AI, etc.
- Format: Table with Question, Answer, Category, Difficulty
- Use: Practice for interviews, learn advanced software topics, or help interviewers prepare questions.

Resume Dataset:

AI-Powered Resume Screening Dataset

- Size: 2,000+ records (Resumes and Job Descriptions)
- Topics: Artificial Intelligence, Resume Parsing, Job Matching, Candidate Ranking, HR Tech.
- Format: CSV file containing Resume Text, Job Description, Similarity Score, and Match Status.
- Use: Train machine learning models for automated recruitment, build resume scanners, or analyze alignment between candidate skills and job requirements.

Screening Dataset:

Resume Dataset

- Size: 960+ resumes

- Topics: Data Science, HR, Banking, Engineering, Information Technology, Healthcare, etc.
- Format: CSV file with Category (Industry/Role) and Resume Text.
- Use: Perform Natural Language Processing (NLP) tasks like document classification, text cleaning,
- and keyword extraction to categorize resumes by professional field.

Implemented(Review)

- ☒ Completed the authentication system
- ☒ Implemented dashboard UI
- ☒ Working job search module
- ☒ Resume optimization
- ☒ Mock interview
- ☒ Profile UI

Conclusion

JobCore aims to streamline the job experience lifecycle by integrating semantic job discovery, ATS-friendly optimization, consent-based automation, and AI-enhanced interview coaching. The initial MVP will prioritize features that benefit users, such as increased match relevance, improved resume performance, expedited application flow, and increased interview preparedness. This product will help job seekers automate the application, preparation, and testing processes more quickly and smoothly, which might increase the chance of getting the job.